DATA SCIENCE

Video Game Sales

University of Tehran
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Scraping & Exploratory Data Analysis (EDA)

phase1

Phase 2

Pipeline & Feature Engineering

Modeling & Prediction

Phase 3

STEP 1

Data Scraping

We used www.kaggle.com for collecting data

This dataset contains video game sales data across different platforms, genres, and regions, making it valuable for various analytical and business use cases.

Features:

Rank - Ranking of overall sales

Name - The games name

Platform - Platform of the games release (PC,PS4)

Year - Year of the game's release

Genre - Genre of the game

Publisher - Publisher of the game

NA_Sales - Sales in North America (in millions)

EU_Sales - Sales in Europe (in millions)

JP_Sales - Sales in Japan (in millions)

Other_Sales - Sales in the rest of the world (in millions)

Global_Sales - Total worldwide sales.

There are 16,598 records

Data Sample

# Rank	Name	Platform	Year	Genre	Publisher	# NA_Sales	# EU_Sales	# JP_Sales	# Other_Sales
Ranking of overall sales	games name	Platform of the games release	the game's release	Genre of the game	Publisher of the game	Sales in North America	Sales in Europe	Sales in Japan	Sales in the rest of the world
1 – 16.6k	11493 unique values	DS 13% PS2 13% Other (12274) 74%	2009 9% 2008 9% Other (13739) 83%	Action 20% Sports 14% Other (10936) 66%	Electronic Arts 8% Activision 6% Other (14272) 86%	0 – 41.5	0 – 29	0 – 10.2	0 – 10.6
1	Wii Sports	Wii	2006	Sports	Nintendo	41.49	29.02	3.77	8.46
2	Super Mario Bros.	NES	1985	Platform	Nintendo	29.08	3.58	6.81	0.77
3	Mario Kart Wii	Wii	2008	Racing	Nintendo	15.85	12.88	3.79	3.31
4	Wii Sports Resort	Wii	2009	Sports	Nintendo	15.75	11.01	3.28	2.96
5	Pokemon Red/Pokemon Blue	GB	1996	Role-Playing	Nintendo	11.27	8.89	10.22	1
6	Tetris	GB	1989	Puzzle	Nintendo	23.2	2.26	4.22	0.58
7	New Super Mario Bros.	DS	2006	Platform	Nintendo	11.38	9.23	6.5	2.9
8	Wii Play	Wii	2006	Misc	Nintendo	14.03	9.2	2.93	2.85
9	New Super Mario Bros. Wii	Wii	2009	Platform	Nintendo	14.59	7.06	4.7	2.26
10	Duck Hunt	NES	1984	Shooter	Nintendo	26.93	0.63	0.28	0.47

STEP 2

Feature Engineering

Transforming Raw Data into Insights

1. Regional Sales Ratios

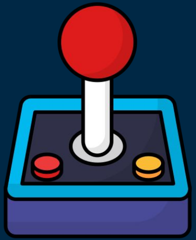
We Created JP_Sales_Ratio, EU_Sales_Ratio, and NA_Sales_Ratio and Each ratio = Regional Sales / Global Sales.it Shows whether a game sold *disproportionately well* in a specific region.

2. Japan Preference Flag

Marked games where JP sales ratio > 75th percentile and Binary Feature: JP_Regional_Preference = 1 if true, else .

3. Platform Sales Index

Measure relative success of a game on its platform .Platform_Sales_Index = Game's sales / Platform's average sales
Insight: Identifies outperformers or underperformers per platform

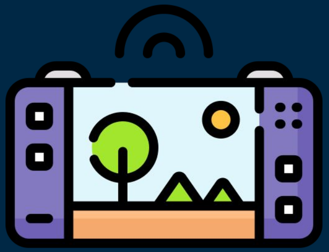


🧹 Null Handling

- **Dropped** rows with missing or invalid Year or Publisher (key fields).
- **Filled** small number of missing numeric values with **mean** or **zero** and Year with KNNImputer.
- Ensured no nulls remained **before encoding** categorical features.
- Goal: Clean dataset for accurate modeling, avoid noise or bias.

📌 Feature Encoding

- Transformed text fields (Genre, Platform, Decade) into numeric form
- Used **One-Hot Encoding** for categorical columns
- Added binary features (e.g., JP_Regional_Preference)
- Created interaction features between Genre & Platform

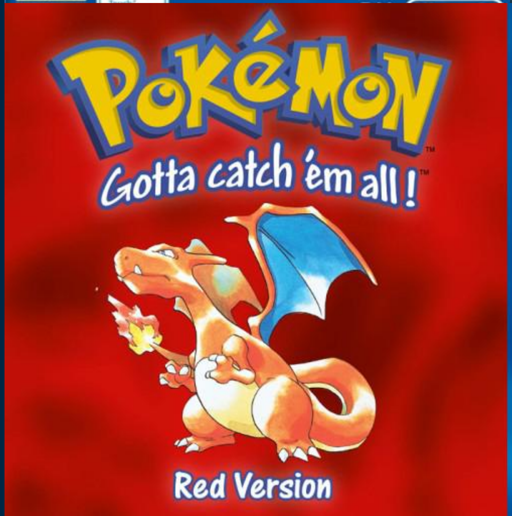


STEP 3

Exploratory Data Analysis

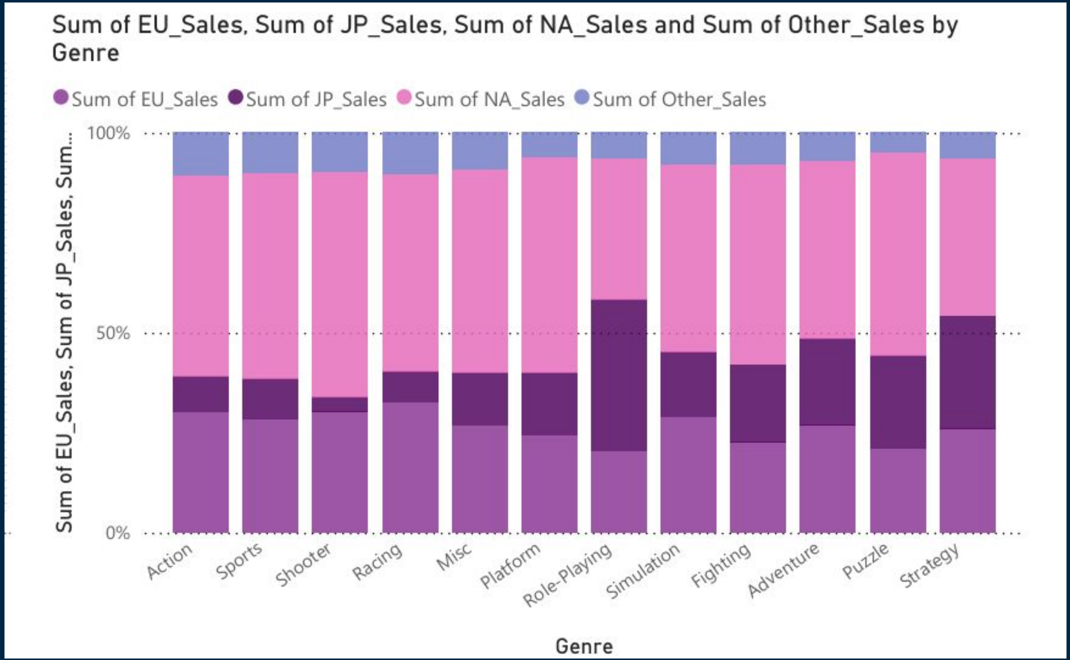
Top 5 Games by Global Sales:

- Wii Sports (Wii) – 82.74M copies
- Super Mario Bros. (NES) – 40.24M copies
- Mario Kart Wii (Wii) – 35.82M copies
- Wii Sports Resort (Wii) – 33.00M copies
- Pokémon Red/Blue (GB) – 31.37M copies



EDA

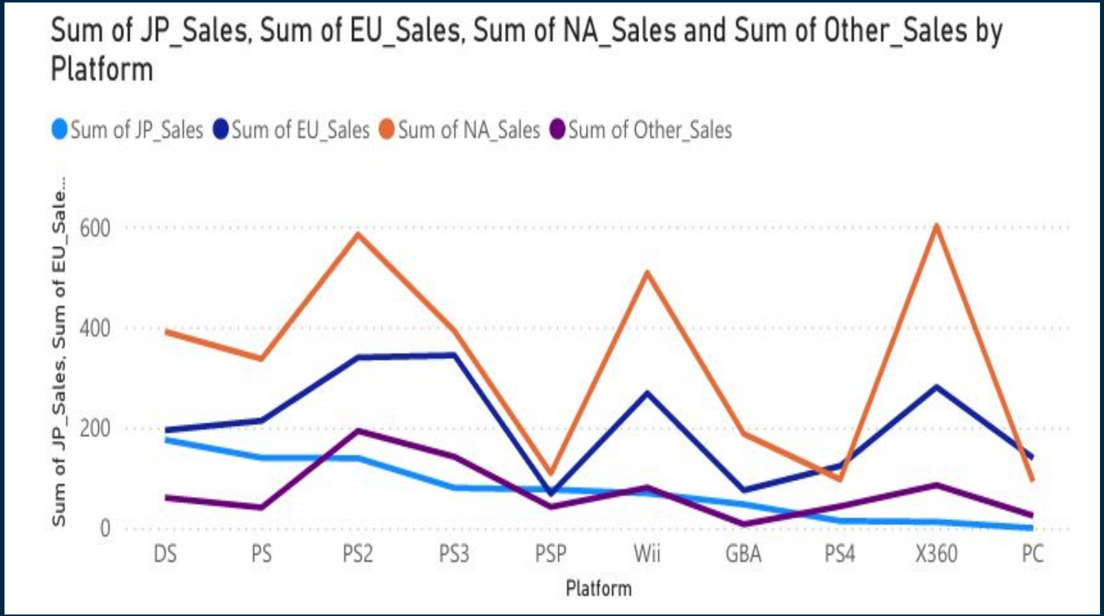
This stacked bar chart illustrates the total video game sales across four major regions—**North America (NA), Europe (EU), Japan (JP), and Other** —grouped by game genre.

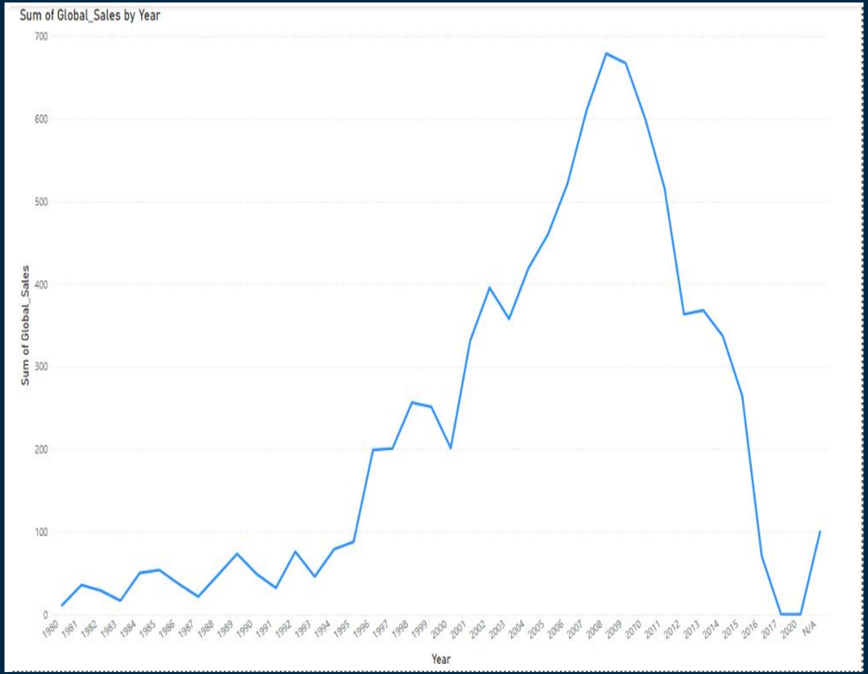


EDA

📈 Line Chart – Regional Sales by Platform

- 🟠 NA sales dominate, especially for **PS2**, **Wii**, **X360**
- 🔵 Japan favors **DS** and **PSP**, but dislikes **X360**/**PC**
- 🟣 PC has lowest sales across all regions
- 🎯 Platforms like **Wii** perform well globally





This line chart displays the trend of total global video game sales by year.

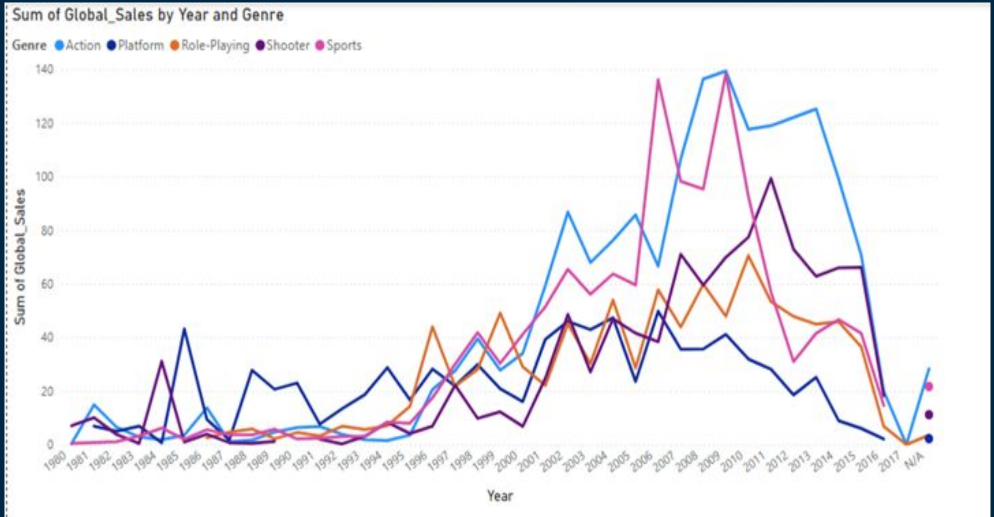
Sales steadily increased from 1995 to 2008, peaking around 2008–2009.

A sharp decline followed in the 2010s.

A possible reason is the rise of digital distribution and free-to-play games, which reduced the need for physical sales tracking. The video game market underwent a major shift after 2008, moving away from physical purchases toward online and free-to-play models.

📌 Key Points:

- **Action** games consistently dominate global sales.
- **Role-Playing** games have lower overall global sales, but still rank 4th globally .
- Sales growth of **Role-Playing** is likely boosted by its popularity in **Japan** .
- Different genres perform differently across regions – regional preferences matter.



Genre	Sum of Global_Sales
Action	1,751.18
Sports	1,330.93
Shooter	1,037.37
Role-Playing	927.37
Platform	831.37
Total	**5,878.22**

[Only NA Sales]	[Only EU Sales]	[Only JP Sales]	[Only Other Sales]
951	546	3137	2

- **Role-Playing** games show very high sales in **Japan** , more than in other regions (3,137).
- **NA** and **EU** regions have more diverse genre sales; **Japan** shows clear concentration in one genre.
- Role-Playing accounts for a **significant share of JP market** , reflecting cultural preference.

STEP 4

Final Steps

Pipeline : Automated Data Preparation

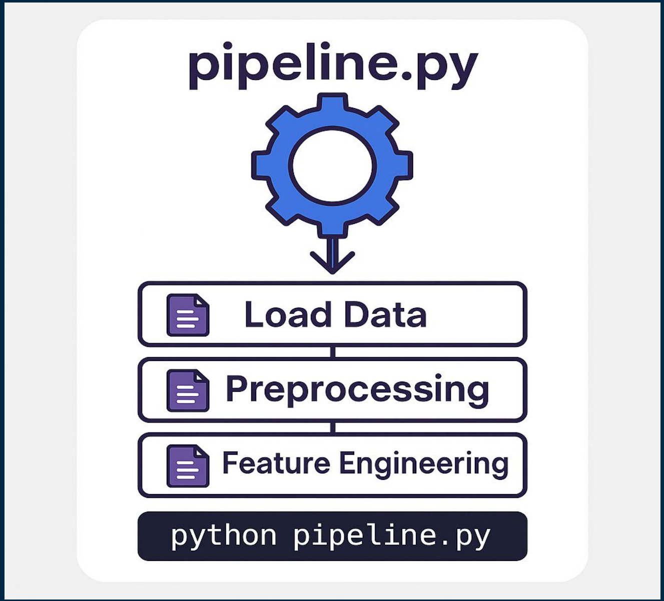
To streamline our workflow, we built an automated pipeline script named pipeline.

📦 Runs all core data scripts in order:

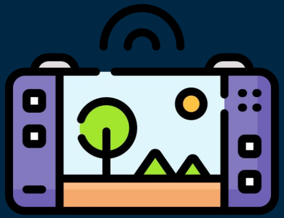
- load_data.py ⟶ loads data from database
- preprocess.py ⟶ cleans and encodes data
- feature_engineering.py ⟶ creates new features

🔁 Ensures repeatability and saves time

✅ One command runs the full pipeline



Automated Hyperparameter Tuning with Optuna



- **Optuna** is an open-source framework for **automated hyperparameter tuning** .
- It uses intelligent search algorithms (like **TPE**) to find the best parameter values.
- Much **faster and more efficient** than manual trial-and-error.
- Boosts model performance with minimal manual effort.
- Easily integrates with most ML frameworks (scikit-learn, PyTorch, XGBoost, etc.)

Best model & why do we choose RMSE?

Model	Best Hyperparameters	RMSE	MSE	MAE	R^2
LinearRegression	{}	0.5649	0.2951	0.2734	0.4271
Ridge	{'alpha': 2547.5521466554087}	0.5553	0.2980	0.2713	0.4417
RandomForest	{'n_estimators': 77, 'max_depth': 20, 'min_samples_split': 2, 'min_samples_leaf': 2}	0.2611	0.0089	0.0181	0.9833
XGBoost	{'n_estimators': 167, 'learning_rate': 0.1494966987593983, 'max_depth': 5, 'subsample': 0.8924239906493234, 'colsample_bytree': 0.714348238249994}	0.1842	0.0057	0.0403	0.9892
NeuralNetwork	{'hidden_layer_sizes': (100,), 'activation': 'tanh', 'max_iter': 577, 'learning_rate_init': 0.00027275613701988256}	0.1837	0.0149	0.0661	0.9721

Best model: NeuralNetwork with RMSE = 0.1837

We choose RMSE as the objective for Optuna optimization because it is smooth, interpretable, and emphasizes large errors – making it ideal for training models to predict game sales accurately.

After comparing multiple models using RMSE, MSE, MAE, and R^2, the Neural Network was selected as the final model due to its lowest RMSE and excellent performance across all metrics.

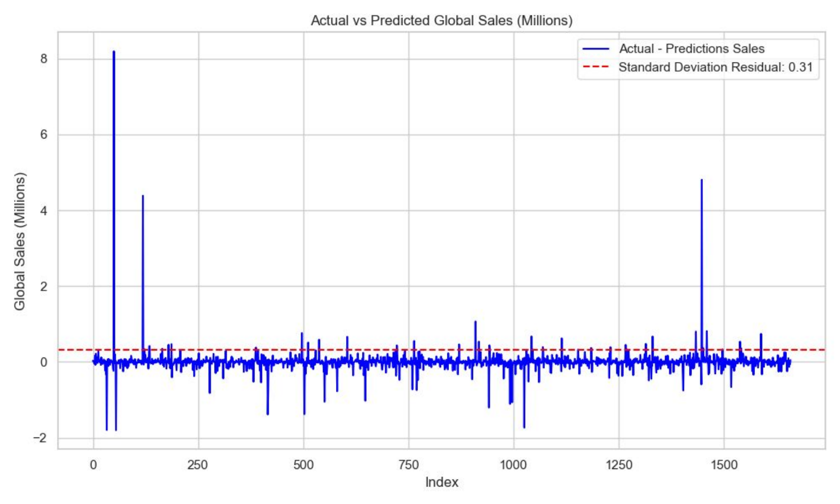
STEP 5

Prediction

Residual Analysis – Actual vs Predicted Global Sales

📌 This chart visualizes the residuals (differences between actual and predicted global sales) across all test samples. The blue line represents the prediction error for each record, while the dashed red line shows the standard deviation of residuals.

🔍 Most prediction errors are tightly centered around **zero**, indicating strong overall model accuracy. A few large spikes (outliers) are visible, showing some games were either **overestimated** or **underestimated** significantly.The **standard deviation of 0.31** confirms that the model is generally consistent, with only minor fluctuations around true values.



Prediction Accuracy – Actual vs Predicted Sales

📌 This scatter plot compares the **actual global sales** (x-axis) to the **predicted global sales** (y-axis) for each game.

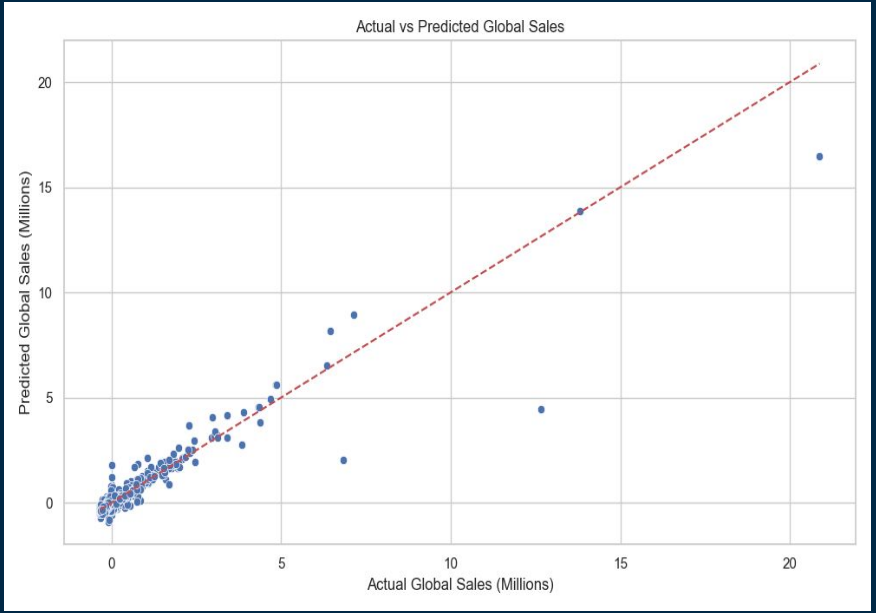
The **red dashed line** represents a perfect prediction line, where predicted values exactly match the actual ones.

Most data points are closely clustered around the line, indicating that the model predicts sales with high accuracy.

A few outliers exist (especially at higher sales values), where predictions deviate from reality.

The model generalizes well for the majority of samples, especially those in the low-to-medium sales range.

💡The regression model performs reliably for most cases, though it tends to underpredict extreme bestsellers.



Potential Uses :

Market Analysis & Trends:

- Identify best-selling genres and platforms over the years.
- Analyze the impact of publishers on sales performance.
- Track sales trends across regions (NA, EU, JP, Other).

Business Strategy & Game Development:

- Help game developers understand what type of games sell best.
- Guide marketing strategies by targeting specific regions with high demand.

Predictive Analytics & Forecasting:

- Build models to predict future game sales based on historical data.
- Identify factors influencing high sales (platform, genre, publisher).

Comparative Analysis:

- Compare sales performance of different gaming platforms (PlayStation, Xbox, Wii).
- Evaluate how different game publishers perform in various markets.

Do you have any questions?

THANKS